How PatchTST-FM makes a forecast

IBM · 385M parameters · Conformer encoder over overlapping patches · zero-shot

1. One series at a time

PatchTST-FM reads only the target's own past. A ? is a value not known yet.

past *future*

Target temperature

Covariates are not used; five metal prices become five separate forecasts.

2. Overlapping patches

The context is scaled by its own mean μ and standard deviation σ. Patches of 16 steps start every 8 steps, so neighbours overlap by half.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

dashed: masked future

Values and mask go through a residual MLP; a learned position embedding is added.

3. Thirty Conformer blocks

Forecasting means filling in the masked future patches. Every part has a LayerNorm in front and a shortcut around it.

½ feed forward (GELU)

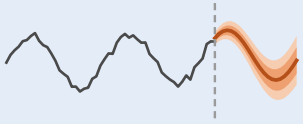
Self-attention: 16 heads, every patch reads every other, past and future

Convolution: depthwise, kernel 5 or 3, smooths across neighbouring patches

½ feed forward (GELU)

4. 99 quantiles, blended

A head turns each future patch into **99 quantiles** (0.01 to 0.99) per step. Overlapping patches are blended with a Hamming window.



median
60% band
80% band

The point forecast is the mean: the area under the quantile curve, not the median.

A wide, bidirectional encoder fills in the future like a gap in the data, in one pass.

30 blocks · width 1,024 · 16 heads · patches of 16 every 8 steps · context up to 8,192 steps

In the notebooks: last 2,048 days or full monthly history · each metal on its own · no covariates · quantiles 0.1 to 0.9 kept